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REVIEW

Neurodevelopmental, Pharmacological and Substance Use Factors in the Association Between ADHD and First-Episode Non-Affective Psychosis: A Systematic Review

Valerio Ricci[1] | Marialuigia Barresi[1] | Gabriele Di Salvo[2] | Giuseppe Maina[1,2]

[1]San Luigi Gonzaga Hospital, University of Turin, Orbassano (TO), Italy | [2]Department of Neurosciences "Rita Levi Montalcini", University of Turin, Torino (TO), Italy

Correspondence: Valerio Ricci (v.ricci@sanluigi.piemonte.it)

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ABSTRACT

Objective: This systematic review investigates the complex relationship between attention deficit hyperactivity disorder (ADHD) and first-episode non-affective psychosis (FEP), focusing on neurodevelopmental, pharmacological and substance use factors that may influence this association.

Methods: Following PRISMA guidelines, we conducted a comprehensive literature search across PubMed, Scopus, Web of Science and PsycINFO databases for studies published between January 2001 and June 2024. We included experimental and observational studies examining ADHD and FEP in participants aged ≥ 16 years. Quality assessment was performed using standardised tools specific to each study design.

Results: From 1243 initially identified records, 13 studies met inclusion criteria. Five studies addressing neurodevelopmental factors revealed shared neurobiological vulnerabilities between ADHD and psychosis, including impaired cortical inhibition, neurophysiological markers and cognitive deficits. Five pharmacological studies demonstrated that whilst stimulant-induced psychosis can occur, the absolute risk appears low, with amphetamines carrying a 1.65-fold higher risk than methylphenidate. Three studies on substance use patterns showed that ADHD patients with FEP had significantly higher rates of substance use disorders, particularly cannabis use, which was associated with earlier psychosis onset and poorer outcomes.

Conclusions: Individuals with ADHD, particularly those with persistent symptoms from childhood and neurocognitive deficits, may have an elevated risk of developing FEP. Substance use appears to be an important mediating factor in this relationship. Regarding pharmacological treatment, recent epidemiological evidence provides a more nuanced perspective on stimulant safety, though the evidence remains mixed and requires further investigation. These findings highlight the importance of comprehensive assessment, substance use prevention and individualised risk stratification when managing ADHD patients at risk for psychosis.

1 | Introduction

The association between attention deficit hyperactivity disorder (ADHD) and psychosis has received increasing attention in psychiatric and neurodevelopmental research. ADHD is a neurodevelopmental disorder classified within major international diagnostic systems, including the DSM-5-TR and ICD-10. Although commonly diagnosed in childhood, ADHD frequently persists into adulthood, affecting approximately 3%–7% of school-aged children (Polanczyk et al. 2015). The disorder is

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primarily defined by deficits across three core domains: inattention, hyperactivity and impulsivity, typically organised into two main symptom clusters. According to the DSM-5-TR, the disorder manifests in different clinical presentations: predominantly inattentive, predominantly hyperactive–impulsive and combined presentation.

Whilst ADHD as a disorder tends to persist over time, the specific manifestation of symptoms evolves during development. Notably, hyperactive–impulsive behaviours generally diminish with age, whereas inattentive symptoms are more likely to persist into adulthood (Carter et al. 2024; Murray et al. 2021; Wang et al. 2024). This developmental trajectory is observed in both male and female patients. ADHD is thought to result from underlying neurobiological dysfunctions, leading to chronic issues with attention, restlessness and impulsivity, which substantially impair daily functioning (Henning et al. 2024; Murray et al. 2018). Adult ADHD prevalence is estimated to be between 4% and 6%. Recent epidemiological data indicate that the male-to-female ratio varies significantly by developmental stage and has narrowed over time due to improved recognition in females. In childhood, clinical samples show ratios of approximately 4:1, whilst community-based studies suggest ratios closer to 2:1 (Martin 2024). In adulthood, this ratio further narrows to approximately 2:1 or approaches 1:1 in community-based epidemiological studies, reflecting the historical underdiagnosis of females during childhood (Martin 2024; Stibbe et al. 2020).

The conceptualisation of schizophrenia as a neurodevelopmental disorder – characterised by early developmental abnormalities that manifest progressively from childhood through adolescence – reveals significant overlap with ADHD in terms of timing, cognitive profile and underlying neurobiological mechanisms (Gogtay and Thompson 2010). Whilst our review focuses specifically on first-episode non-affective psychosis (FEP), we acknowledge that schizophrenia represents a primary example within this diagnostic category. Shared features between the two conditions are evident in childhood, particularly in the form of cognitive alterations that emerge during school age. These cognitive dysfunctions include deficits in sustained attention, executive functions (planning, response inhibition, cognitive flexibility), working memory and information processing speed. In vulnerable individuals, there may be a gradual worsening of subthreshold symptoms that can eventually culminate in full-blown psychosis (Cornblatt et al. 2003).

From a clinical perspective, both ADHD and schizophrenia present specific overlapping characteristics that distinguish their relationship from other psychiatric comorbidities. These include shared deficits in executive functioning domains (working memory, cognitive flexibility, inhibitory control), similar patterns of attention and information processing abnormalities, comparable impairments in social cognition and theory of mind, and overlapping neurobiological dysfunction in dopaminergic and frontostriatal circuits (Hasan et al. 2013). Additionally, both disorders frequently show abnormalities in impulsivity and inhibitory control domains, which may contribute to maladaptive behaviours and functional difficulties (Lim et al. 2018; Vande Voort et al. 2014).

The relationship between ADHD and psychosis is complex, with growing evidence indicating shared neurobiological pathways, genetic susceptibilities and environmental risk factors (Mahrous et al. 2024; Pingault et al. 2015; Zheng et al. 2020). Specifically, schizophrenia shares numerous genetic and environmental risk factors with ADHD, including genetic variants involved in dopaminergic neurotransmission and pre- and perinatal complications (Lim et al. 2018). Both disorders exhibit common clinical characteristics, including a male predominance, high comorbidity with substance abuse, and challenges in emotional regulation and social relationships. Furthermore, individuals with ADHD are at increased risk of developing comorbid conditions, such as substance use disorders (SUDs), which may further contribute to the progression toward psychosis (Katz et al. 2009; Zucker et al. 2002).

Recent studies have also highlighted that patients with FEP and a history of ADHD (FEP-ADHD+) may represent a distinct subgroup with a worse prognosis, characterised by earlier onset of psychosis, more pronounced negative symptoms and greater cognitive deficits (Sanchez-Gistau et al. 2020). The study by Hollis et al. (2019) published in *Lancet Psychiatry* further emphasised the importance of this association, demonstrating a significantly increased risk of developing psychotic disorders in patients with ADHD.

This systematic review aims to investigate the complex connection between ADHD and the onset of FEP, focusing on identifying the key factors involved in the development of psychotic symptoms, including the role of pharmacological treatment. Amongst the specific objectives, we intend to analyse which psychopathological factors may predict the development of psychosis in ADHD patients, examine the evidence regarding risks associated with pharmacological treatments for ADHD (including methylphenidate and amphetamines), and evaluate the clinical implications for managing patients with comorbidity.

Through an extensive review of the literature, including experimental studies, observational studies, case reports and case series, this study seeks to provide a comprehensive analysis of the current evidence, emphasising shared neurodevelopmental disruptions and potential pathogenic mechanisms linking ADHD and psychosis.

2 | Methods

Following the methodology proposed by Moher et al. (2009) and adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, we conducted a systematic review to thoroughly examine the relationship between ADHD and first-episode psychosis. This process involved defining precise research objectives, implementing a comprehensive search strategy, applying rigorous study selection criteria, conducting thorough data extraction, assessing the quality of included studies and synthesising the findings in a structured narrative format. This methodical approach ensured transparency and reproducibility whilst allowing for a comprehensive analysis of the current evidence base.

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2.1 | Research Questions and Conceptual Framework

Prior to conducting the literature search, we developed a comprehensive conceptual framework based on existing evidence that suggested three key domains relevant to the relationship between ADHD and first-episode psychosis: neurodevelopmental factors, pharmacological impacts (particularly stimulant medications) and substance use patterns. Our primary research aims were to identify the neurodevelopmental factors in ADHD individuals associated with increased risk of developing first-episode psychosis, to evaluate the impact of pharmacological treatments for ADHD (particularly methylphenidate) on psychosis risk, and to examine how substance use mediates the relationship between ADHD and the development of first-episode psychosis. These research objectives guided our systematic approach to literature identification and evidence synthesis.

2.2 | Development of Conceptual Framework and Identification of Thematic Areas

At the beginning of our review process, we conducted a preliminary literature consultation and meetings with clinical experts to identify the main thematic areas connecting ADHD to first-episode psychosis. This systematic process led to the identification of three primary domains: neurodevelopmental factors, pharmacological impacts and substance use patterns. These thematic areas were not arbitrarily determined but emerged from: (i) a preliminary literature search that identified these themes as predominant; (ii) consultations with clinical experts in the field of neurodevelopment and FEP; (iii) analysis of existing theoretical models linking ADHD to psychosis.

This thematic structure guided our search strategy and the organisation of findings, ensuring a systematic and evidence-based approach.

2.3 | Search Strategy and Information Sources

A structured literature search was conducted using multiple electronic databases: PubMed (via MEDLINE), Scopus, Web of Science and PsycINFO on 21 June 2024, covering the period from January 2001 onwards. Additionally, we reviewed reference lists of included articles for potentially relevant studies not captured in the database searches.

The search strategy was designed to identify literature at the intersection of ADHD and first-episode psychosis, whilst also capturing studies related to our three predetermined focus areas. The primary search string included:

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('first episode non-affective psychosis' OR 'first episode psychosis' OR 'FEP' OR 'first episode schizophrenia' OR 'schizophrenia prodromal phase' OR 'schizophrenia prodromal symptoms' OR 'psychosis').
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AND

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('ADHD' OR 'attention-deficit hyperactivity disorder' OR 'attention-deficit spectrum disorder' OR 'hyperkinetic disorder').
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This primary search string was then combined with additional search terms for each focus area:

- For neurodevelopmental factors:

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AND ('neurodevelopment' OR 'cognitive deficit*' OR 'executive function*' OR 'intellectual disability' OR 'autism' OR 'developmental delay*' OR '7mmerik7logy*' OR 'brain development').
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- For pharmacological impacts:

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AND ('methylphenidate' OR 'stimulant*' OR 'amphetamine*' OR 'psychostimulant*' OR 'ADHD medication*' OR 'drug-induced psychosis' OR 'atomoxetine' OR 'pharmacotherapy' OR 'drug safety').
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- For substance use:

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AND ('substance use' OR 'substance abuse' OR 'addiction' OR 'cannabis' OR 'marijuana' OR 'alcohol' OR 'tobacco' OR 'cocaine' OR 'stimulant abuse' OR 'comorbidity' OR 'dual diagnosis').
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2.4 | Eligibility Criteria

Studies were considered eligible if they met the following inclusion criteria: included patients aged 16 and older; applied diagnostic criteria for ADHD in accordance with DSM-IV, DSM-5 or ICD-10/11 guidelines; applied diagnostic criteria for FEP (excluding bipolar disorder, schizoaffective disorder and psychotic depression) in accordance with standardised diagnostic systems; addressed at least one of our three focus areas: neurodevelopmental factors, pharmacological impacts or substance use patterns; original research articles (experimental, observational, post-marketing surveillance reports, case reports and case series); full-text articles available in English.

Studies were excluded if they met any of the following criteria: involved patients with pre-existing diagnoses of schizophrenia or other disorders within the schizophrenia spectrum prior to study initiation; involved patients with personality disorders or neurocognitive impairments such as dementia as primary diagnoses; involved patients with neurological or infectious conditions as primary diagnoses; non-original research (e.g., reviews, commentaries, editorials, book chapters and letters to the editor); non-full-text articles (e.g., meeting abstracts); articles not in English; research incorporating animal or in vitro experiments only. Regarding case reports, our approach evolved during the review process. Whilst initially included for their detailed clinical observations,

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we recognised their methodological limitations compared to observational and experimental studies. Consequently, we adopted a more nuanced strategy, treating case reports as supplementary rather than primary evidence. In our analysis, we carefully distinguished between conclusions derived from case reports and those supported by more robust study designs. We explicitly acknowledged the limitations of case reports in establishing causal relationships, instead utilising them primarily to illustrate potential mechanisms or rare effects that merit further investigation through more rigorous methodologies. This reassessment has led to a clearer hierarchy of evidence in our analysis, with priority given to observational and experimental studies.

2.5 | Study Selection Process

The study selection process followed a two-stage approach. In the first stage, three investigators (V.R., M.B. and G.D.S.) independently screened titles and abstracts against the eligibility criteria. In the second stage, the full texts of potentially eligible studies were retrieved and independently assessed by two investigators (V.R. and M.B.). Any disagreements were resolved through discussion with a third investigator (G.M.) serving as arbiter. The literature search and selection process were conducted by three investigators under the guidance of two supervisors. This involved a two-stage independent review by the investigators, followed by a collaborative cross-check to ensure consistency and thoroughness.

2.6 | Data Extraction and Categorisation

The following parameters were extracted from each eligible article: first author, publication year, country, study design, sample size, participant demographics, diagnostic methods used, measurements/outcomes and key findings related to our three focus areas.

The primary outcome was to identify which psychopathological factors are associated with the development of first-episode psychosis in individuals with ADHD. Data extraction specifically focused on evidence related to: (i) neurodevelopmental factors (including cognitive deficits, neurophysiological markers and comorbid neurodevelopmental disorders); (ii) pharmacological impacts (particularly methylphenidate and other stimulant medications); (iii) substance use patterns (with special attention to cannabis and other substances known to potentially trigger psychosis).

2.7 | Quality Assessment

We conducted a formal quality assessment for all included studies using standardised tools specific to each study design: (i) for observational studies: Newcastle–Ottawa Scale (NOS); (ii) for randomised controlled trials: Cochrane Risk of Bias Tool 2.0; (iii) for case series: Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Case Series; (iv) for case reports: Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Case Reports. Detailed quality assessment scores for individual items are provided in Tables S1–S6.

Two reviewers independently assessed each study, with a third reviewer resolving any disagreements. Quality scores were categorised as high, moderate or low, and these levels of evidence were explicitly considered in the synthesis of evidence and conclusions drawn. Studies with higher levels of evidence received greater weight in the overall analysis. A complete table of quality scores is included in Tables S1–S6.

2.8 | Ethical Considerations

This review article is based exclusively on previously published data and studies and did not involve the collection of any primary data from human or animal subjects. Therefore, ethical approval was not required for this work.

3 | Results

A total of 1243 articles were identified across all databases (538 from PubMed, 412 from Scopus, 175 from Web of Science and 118 from PsycINFO). After removing 243 duplicates, 1000 titles and abstracts were screened, resulting in 82 articles for full-text review. Of these, 13 were ultimately selected based on our predefined criteria. Exclusion of the other 69 articles was due to irrelevance to our specific research questions (42) or non-compliance with selection criteria, including reviews, letters to the editor, commentaries, book chapters and non-English papers (27 articles). The detailed selection process is presented in Figure 1, following the PRISMA flow diagram format.

The relatively small number of included studies reflects the highly specific focus of our review on the intersection between ADHD and first-episode psychosis, which represents an emerging research area with limited dedicated literature. This specificity was intentional to ensure direct relevance to our research questions, though it highlights an important gap warranting future investigation.

These 13 studies selected comprise: five studies on neurodevelopmental factors (one cross-sectional study, three case–control studies and one retrospective cohort study), five studies on pharmacological effects (one RCT, one retrospective cohort study, one retrospective case–control study, one case series and one case report) and three studies on substance use (two prospective cohort studies and one cross-sectional study).

3.1 | Neurodevelopmental Factors

A growing body of literature investigates the intersection between neurodevelopmental disorders such as ADHD and the onset of FEP. Amongst the primary neurodevelopmental factors identified are cognitive delays and other disorders like autism and intellectual disability. Strålin and Hetta (2019) provide significant insights into this area by demonstrating how comorbidity between ADHD, autism and intellectual disability influences the clinical course of FEP. Specifically, they found that patients with autism or intellectual disability, when coupled with psychosis, require longer treatment durations and higher doses of antipsychotic medication compared to other FEP patients.

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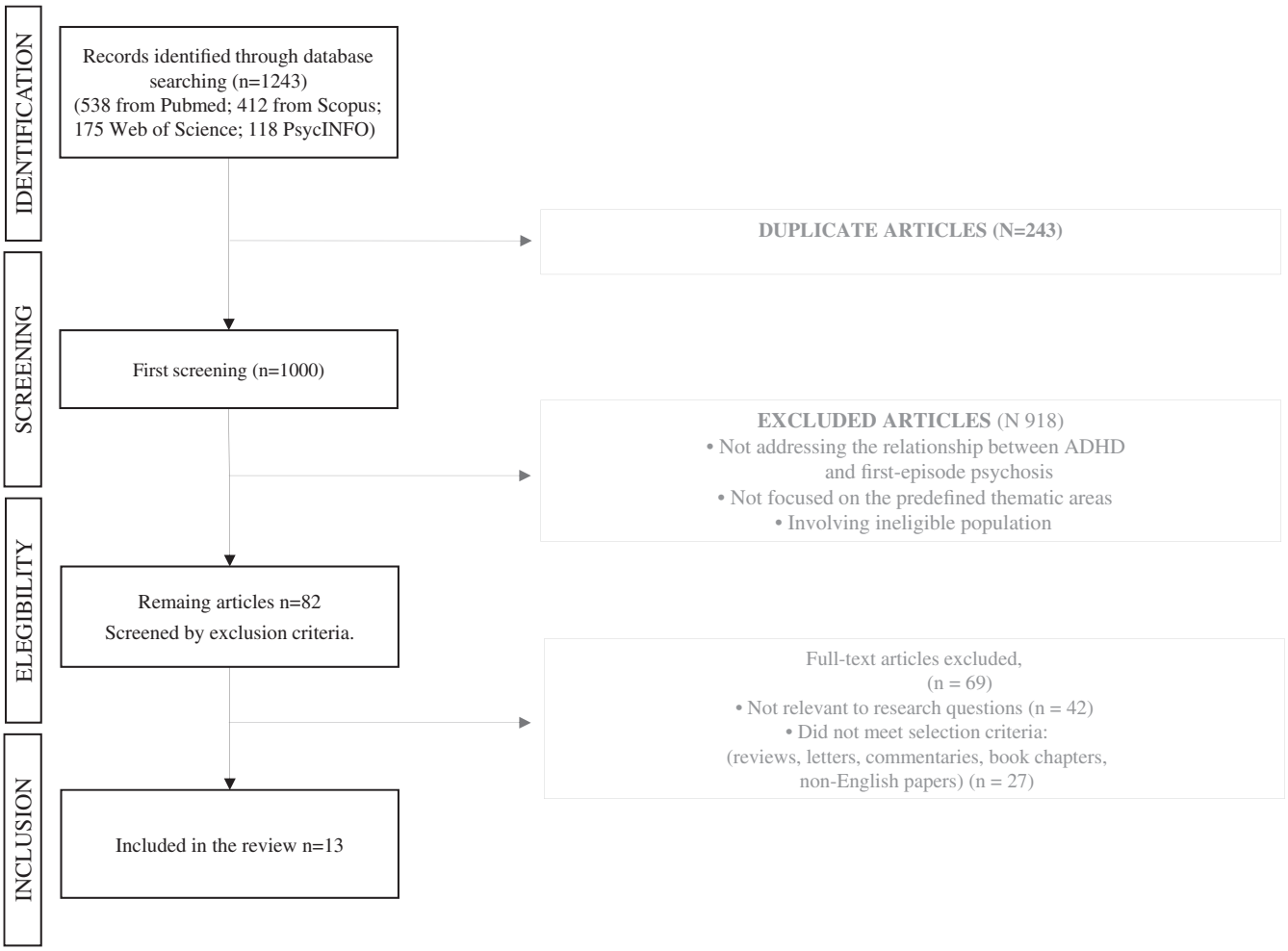


FIGURE 1 | Flow chart of study search and selection process.

Conversely, those with ADHD are at a higher risk for SUDs and self-harm, both before and after the first psychosis hospitalisation (Strålin and Hetta 2019).

Rydkjær et al. (2017) advance this area of research by investigating neurophysiological markers, specifically mismatch negativity (MMN) and P3a amplitude, in adolescents with FEP and ADHD. Their findings indicate that adolescents with FEP demonstrate significant pre-attentive deficits, reflected in reduced MMN amplitudes, when compared to healthy controls. The ADHD group showed intermediate deficits, with MMN amplitudes falling between those of the FEP group and healthy controls, although not significantly different from either. These results suggest a potential overlap in neurodevelopmental abnormalities between ADHD and FEP, possibly pointing to shared neurobiological vulnerabilities (Rydkjær et al. 2017).

Similarly, Sanchez-Gistau et al. (2023) compared cognitive biases in 121 first-episode psychosis patients with and without ADHD versus healthy controls. Results showed FEP patients with childhood ADHD (FEP-ADHD+) had significantly more cognitive biases than those without ADHD (FEP-ADHD−) and healthy controls, particularly in 'Intentionalising' (attributing intentions to others) and 'Emotional Reasoning' (using emotions as facts). Cognitive biases correlated with positive psychotic symptoms in both FEP groups, but with depression only in FEP-ADHD− and with poor functioning only in FEP-ADHD+. The findings suggest FEP patients with comorbid ADHD represent a particularly vulnerable subgroup requiring targeted metacognitive interventions to address their heightened tendency toward biased interpretation of social information.

Peralta et al. (2011) examined the relationship between childhood ADHD symptoms and first-episode schizophrenia-spectrum psychosis, providing support for a neurodevelopmental model. Their findings showed that the severity of childhood ADHD symptoms was significantly related to obstetric complications and neurodevelopmental delays. This indicates that ADHD symptoms in individuals with schizophrenia-spectrum disorders may be part of a broader neurodevelopmental dysfunction, rather than independently contributing to the clinical expression of psychosis. These results highlight shared neurodevelopmental risk factors between ADHD and schizophrenia-spectrum disorders, supporting the hypothesis that ADHD may represent an early manifestation of the underlying neurodevelopmental abnormalities that predispose individuals to psychosis (Peralta et al. 2011).

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Adding to this body of evidence, Hasan et al. (2013) investigated cortical excitability in FEP and ADHD patients using transcranial magnetic stimulation (TMS) to explore whether there are shared neurophysiological pathways between the two disorders. Their findings revealed a similar pattern of impaired cortical inhibition and hemispheric imbalance of excitability in both FEP and ADHD patients, suggesting that these two disorders may share common pathophysiological mechanisms, particularly related to GABAergic and dopaminergic transmission. These results highlight potential neurobiological overlaps between ADHD and psychosis, further supporting the neurodevelopmental hypothesis of comorbidity between these conditions. The main findings related to neurodevelopmental factors are summarised in Table 1.

3.2 | Methylphenidate Therapy

Another critical aspect of the ADHD-FEP relationship concerns the use of stimulant medications like methylphenidate in ADHD treatment, which may trigger psychosis in some patients. Moran et al. (2019) explored the differential risk of psychosis onset between methylphenidate and amphetamine in ADHD patients. Their findings revealed that amphetamine use was associated with a significantly higher risk of inducing psychosis compared to methylphenidate, with approximately 1 in 660 patients experiencing psychosis. The risk was 1.65 times higher for amphetamine users compared to those on methylphenidate, a finding that may be explained by amphetamine's more potent release of dopamine, which closely mimics the dopaminergic dysfunction seen in primary psychotic disorders (PPDs) (Moran et al. 2019).

The study by Takahashi et al. (2014) was a randomised, double-blind, placebo-controlled, parallel-group trial conducted at 39 sites in Japan. It evaluated the efficacy and safety of osmotic-controlled release oral delivery system (OROS) methylphenidate (MPH) in adults with ADHD. A total of 284 adults were randomised to receive either OROS MPH or placebo. During a 4-week titration period, patients were started on 18 mg/day of OROS MPH, with weekly increments of 18 mg to reach an individualised dose, up to a maximum of 72 mg/day. The primary endpoint was assessed after a subsequent 4-week efficacy assessment period. In this study, one patient in the OROS MPH group developed a psychotic disorder, which was classified as a serious treatment-emergent adverse event (TEAE). The investigators determined that the psychotic episode was probably related to the methylphenidate treatment. This highlights the importance of monitoring patients closely during stimulant treatment, especially those with potential predispositions to psychosis (Takahashi et al. 2014).

Three cases were reported in which patients, aged 29, 38 and 45, with no prior history of psychotic symptoms, developed such symptoms after taking standard doses of methylphenidate. Once methylphenidate was discontinued and antipsychotic treatment was initiated, their psychotic symptoms resolved within 2–21 days (Kraemer et al. 2010).

Auger et al. (2005) conducted a retrospective case–control study at a sleep centre, involving 116 patients with narcolepsy or idiopathic hypersomnia. They compared a high-dose stimulant group (≥ 120% of the maximum recommended stimulant dose) to a standard-dose group. The high-dose group showed a significantly higher prevalence of psychosis (24% vs. 5% in the standard-dose group), with an odds ratio of 12.0. Notably, six patients in the high-dose group developed psychosis on methylphenidate at a median dose of 170 mg/day (range 120–500 mg), and their symptoms improved with dose reduction to a median of 80 mg/day (Auger et al. 2005).

Additionally, Ghadrdan et al. (2018) reported a rare case of methylphenidate-induced psychosis in a 65-year-old female with ADHD. The patient, who was consuming three to four methylphenidate tablets daily, developed psychotic symptoms which resolved after discontinuation of the stimulant and initiation of antipsychotic medication. This case highlights that psychosis induced by methylphenidate, whilst uncommon, can occur in both young and elderly populations and that dose mismanagement may play a role in precipitating psychotic episodes (Ghadrdan et al. 2018). The main findings related to neurodevelopmental factors are summarised in Table 2.

3.3 | Substance Use in First-Episode Psychosis With ADHD

The relationship between ADHD and substance use in patients with FEP is well-established, with multiple studies demonstrating that ADHD serves as a significant predictor of substance abuse, particularly cannabis use. Sanchez-Gistau et al. (2020) examined 133 FEP patients, with approximately 25% presenting childhood ADHD (c-ADHD). Results showed that the FEP-ADHD+ group had significantly worse cognitive performance, particularly in attention, processing speed and executive function compared to FEP-ADHD− and healthy controls. Clinically, the FEP-ADHD+ group had fewer years of education, higher rates of tobacco and cannabis use, and required higher antipsychotic doses to achieve clinical response. Whilst the two FEP groups did not differ in psychopathology severity, there was a gradient of cognitive severity increasing from healthy controls to FEP-ADHD− to FEP-ADHD+. The study concludes that FEP-ADHD+ represents a subgroup with worse prognosis than FEP-ADHD.

In a prospective observational study involving 179 first-episode psychosis (FEP) patients, with 15% having a history of childhood ADHD, it was found that FEP patients with ADHD had poorer premorbid functioning, lower educational achievement and an earlier onset of psychosis compared to those without ADHD. Additionally, the FEP-ADHD group demonstrated worse clinical outcomes, with more severe positive and negative symptoms, as well as poorer social and occupational functioning over time. Notably, the FEP-ADHD group had a higher prevalence of SUDs (75% vs. 56%), particularly involving tobacco and cannabis, which likely contributed to their increased risk of psychosis and poorer treatment response (Rho et al. 2015).

Cassidy et al. (2011), in a prospective observational study, explore the relationship between childhood ADHD symptoms and subsequent cannabis use in 75 patients with FEP. Using the Child Behaviour Checklist (CBCL), the study assessed childhood symptoms of hyperactivity–inattention and their

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TABLE 1 | Main findings related to neurodevelopmental disorders on psychosis risk in patients with ADHD.

Author	Year	Study design	Sample	Key measures	Main findings	Quality
Peralta et al.	2011	Cross-sectional	84 first-episode psychosis patients (64.3% male)	PAS, CASH, CBCL	Childhood ADHD symptoms correlated with obstetric complications ($r = 0.36$) and neurodevelopmental delays ($r = 0.41$), supporting shared risk model.	High
Hasan et al.	2013	Case–control	20 FEP, 20 ADHD, 20 controls	TMS measures, PANSS, ADHD checklist	Both FEP and ADHD showed impaired cortical inhibition, suggesting shared GABAergic and dopaminergic dysfunction.	Moderate
Rydkjær et al.	2017	Case–control	30 FEP adolescents, 30 ADHD, 30 controls	MMN and P3a amplitude (EEG)	FEP patients showed reduced MMN amplitudes; ADHD group showed intermediate deficits between FEP and controls.	High
Strålin and Hetta	2019	Retrospective cohort	792 FEP patients, including 55 with ADHD	Clinical assessments, treatment duration, medication doses	FEP + ADHD patients had higher rates of substance use (38.2% vs. 12.5%) and self-harm (36.4% vs. 10.4%) than FEP-only patients.	High
Sanchez-Gistau et al.	2023	Cross-sectional	121 FEP patients, 31 (25.6%) with childhood ADHD	DIVA, PANSS, BACS	FEP-ADHD+ showed greater cognitive impairment, higher substance use and required higher antipsychotic doses than FEP-ADHD− patients.	High

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TABLE 2 | Main findings related to stimulant-induced psychosis risk in ADHD.

Author	Year	Study design	Sample	Intervention	Main findings	Quality
Auger et al.	2005	Retrospective case–control	116 patients with narcolepsy/hypersomnia	High-dose vs. standard-dose stimulants	High-dose group showed higher psychosis prevalence (24% vs. 5%, OR = 12.0); symptoms improved with dose reduction.	Moderate
Kraemer et al.	2010	Case series	3 adult ADHD patients (ages 29–45)	Methylphenidate 30–40 mg/day	All developed psychotic symptoms on standard doses; resolved within 2–21 days after discontinuation.	Low
Takahashi et al.	2014	RCT	284 adults with ADHD (142 MPH, 142 placebo)	OROS methylphenidate (18–72 mg/day) for 8 weeks	One patient (0.7%) in MPH group developed psychosis classified as treatment-related serious adverse event.	High
Moran et al.	2019	Retrospective cohort	337 919 ADHD patients (13–25 years)	Methylphenidate (n = 110 923) vs. amphetamine (n = 226 996)	Amphetamine associated with higher psychosis risk (HR = 1.65); incidence: 1.78 vs. 2.83 per 1000 person-years.	High
Ghadrdan et al.	2018	Case report	65-year-old female with ADHD	Methylphenidate 30–40 mg/day for 2 weeks	Developed persecutory delusions; resolved after discontinuation and risperidone treatment.	Low

ability to predict cannabis use following a FEP diagnosis. The findings revealed that early life symptoms of inattention and hyperactivity significantly predicted lifetime cannabis use disorder and an inability to maintain abstinence from cannabis during treatment. Specifically, the inattention symptom alone was a stronger predictor of ongoing cannabis use, even after controlling for other factors like conduct disorder (CD) and oppositional defiant disorder (ODD). The study highlights that inattention–hyperactivity in childhood may play a crucial role in predisposing individuals to cannabis use in the context of psychosis, which could, in turn, exacerbate psychotic symptoms (Cassidy et al. 2011).

Overall, the association between ADHD and substance use in FEP patients is clear, with cannabis use being a particularly prominent issue. These findings highlight the importance of early identification and management of substance use in ADHD patients with psychosis to improve clinical outcomes and prevent further complications. The main findings related to substance use and its influence on psychosis risk are summarised in Table 3.

3.4 | Quality Assessment Results

The quality assessment revealed generally good methodological rigour across included studies. Of the 13 studies, 8 (61.5%) were rated as high quality, 3 (23.1%) as moderate quality and 2 (15.4%) as low quality. The single RCT demonstrated low risk of bias across all assessed domains. Cohort and case–control studies scored well on the Newcastle–Ottawa Scale (median score 7/9), with good representativeness and adequate follow-up. Case studies, whilst methodologically limited by design, provided valuable clinical insights with adequate reporting quality.

Higher quality studies were given greater weight in evidence synthesis, particularly the large epidemiological studies (Hollis et al. 2019; Moran et al. 2019) which informed our primary conclusions regarding pharmacological safety.

4 | Discussion

Based on the data obtained thus far, we have identified three key areas concerning the onset of non-affective psychosis in individuals with ADHD. Our findings suggest that the first episode of psychosis appears to occur more frequently in ADHD patients with neurocognitive deficits or delays, particularly those with persistent ADHD symptoms from childhood (referred to as childhood-onset ADHD that continues into adolescence and adulthood). Furthermore, ADHD patients with a history of substance use appear to have an increased susceptibility to psychotic episodes. The relationship between pharmacological treatment of ADHD and psychosis risk requires nuanced interpretation.

Regarding the first point, evidence suggests potential shared neurobiological abnormalities, as well as genetic and environmental risk factors, between ADHD and schizophrenic psychosis, pointing toward a possible common pathogenetic

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TABLE 3 | Main findings related to substance use and its influence on psychosis risk in patients with ADHD.

Author	Year	Study design	Sample	Key measures	Main findings	Quality
Cassidy et al.	2011	Prospective cohort	75 FEP patients (70.7% male)	CBCL, SCID, Timeline Follow-Back	Childhood inattention-hyperactivity predicted cannabis use disorder (OR = 2.8) and inability to maintain abstinence (OR = 7.1).	High
Rho et al.	2015	Prospective cohort	179 FEP patients, 27 (15.1%) with ADHD history	K-SADS, PANSS, CDSS, SOFAS	FEP-ADHD had earlier psychosis onset (21.1 vs. 23.6 years), higher SUD prevalence (75% vs. 56%), more severe symptoms and poorer outcomes.	High
Sanchez-Gistau et al.	2020	Cross-sectional	133 FEP patients, 33 (24.8%) with childhood ADHD	DIVA, MCCB, PANSS	FEP-ADHD+ showed higher rates of tobacco and cannabis use, more cognitive impairments (especially in attention, processing speed and executive function), fewer years of education and required higher antipsychotic doses.	High

mechanism (Colaizzi et al. 2023; Erlenmeyer-Kimling 2001; Frodl 2010; Muhammad et al. 2023; van Emmerik-van Oortmerssen et al. 2012). ADHD, as a neurodevelopmental disorder, may serve as a risk factor for developing psychosis in adulthood. This risk may be mediated by several factors observed in both conditions, including genetic variants affecting dopaminergic pathways, neurodevelopmental abnormalities, impaired executive functioning and structural brain alterations. Additionally, comorbid conditions such as CD and intellectual disability may contribute to this risk, suggesting that ADHD may impact neurodevelopment in ways that potentially increase vulnerability to psychosis (Dennis et al. 2004; Kousha et al. 2012).

Recent broader studies beyond our review have also identified shared genetic risk loci between ADHD and schizophrenia, particularly in genes involved in dopamine regulation and neurodevelopment (Demontis et al. 2019; Faraone and Larsson 2019). These genetic overlaps provide additional support for neurodevelopmental connections between these conditions.

Supporting these theories, schizophrenia is increasingly conceptualised as a neurodevelopmental disorder, with brain abnormalities and prodromal symptoms manifesting during childhood and adolescence, well before the onset of psychosis. Risk factors that may be shared with ADHD include obstetric complications, early neurodevelopmental disruptions, alterations in dopamine system functioning, genetic predisposition and early life adversities. These factors may interact with environmental stressors, potentially influencing neurodevelopment and the transition to psychosis. Additionally, inattention has been associated with the development of psychosis during late adolescence and adulthood in some studies (Kraemer et al. 2010).

The extensive discussion of SUDs in our review is warranted by evidence indicating that SUD may represent an important mediating factor in the relationship between ADHD and psychosis. Adolescents with SUD appear to have higher rates of ADHD compared to the general population (Colaizzi et al. 2023; Frodl 2010; Muhammad et al. 2023; van Emmerik-van Oortmerssen et al. 2012). For instance, an NIH study found ADHD in 38% of adolescents with cannabis use disorder (Dennis et al. 2004), whilst a meta-analysis by van Emmerik-van Oortmerssen et al. reported ADHD in 23% of young adults seeking treatment for substance abuse (van Emmerik-van Oortmerssen et al. 2012). The International ADHD in Substance Use Disorders Prevalence (IASP) study, involving 3558 participants from 10 countries, reported a 40% prevalence of ADHD (van de Glind et al. 2013).

Research suggests that children with ADHD may be at increased risk of developing SUD, with potentially higher risk if they have comorbid CD (Ercan et al. 2003). A meta-analysis by Charach et al. found ADHD youth were 2.4 times more likely to smoke and 1.5 times more likely to develop SUD (Charach et al. 2011; Lee 2011; Margherio et al. 2024), with Groenman et al. reporting similar findings in a 4-year follow-up (Groenman et al. 2013). Some studies indicate that ADHD youth may develop SUD earlier and progress more rapidly (King et al. 1999; Kousha et al. 2012; Riggs 1998; Riggs et al. 2004), particularly with comorbid CD (González et al. 2020; Mustonen et al. 2023).

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Research indicates that approximately 50% of individuals with mental disorders, particularly schizophrenia, have a co-occurring SUD (Ricci, Ceci, et al. 2021), higher than rates observed in the general population. Substance-induced psychosis (SIP) has been reported in 7%–25% of first-episode psychosis cases. A study in Scandinavia found the annual incidence of SIP was 6.5 per 100 000, compared to 9.7 amongst those with a PPD and SUD, and 24.1 in PPD alone (Ricci et al. 2023; Ricci, Martinotti, et al. 2021). Studies suggest ADHD patients with SUD may face an increased risk of psychosis, possibly because both conditions involve dopaminergic system alterations, and early substance use, especially cannabis and tobacco, has been associated with psychosis in some individuals (Ghadrdan et al. 2018; Rho et al. 2015; Sanchez-Gistau et al. 2020).

Regarding pharmacological treatment of ADHD, the evidence warrants careful interpretation. The well-designed study by Hollis et al. (2019) challenges concerns about methylphenidate and psychosis risk. Their data showed 'no evidence of an increase in the immediate risk of psychotic events when comparing the 12-week periods before and after initiation of methylphenidate treatment, neither for those with or without a history of psychosis'. More notably, 1 year after starting methylphenidate treatment, they observed 'a 36% reduction in the incidence of psychotic events in those with a history of psychosis and an 18% reduction in incidence in those without a history of psychosis relative to the period immediately before initiation of methylphenidate treatment'. These findings contribute important evidence that challenges concern about methylphenidate use in individuals with or without a history of psychosis.

This perspective is further supported by larger meta-analyses not included in our primary review. A systematic review and meta-analysis by Cortese et al. (2018) examining 10 randomised controlled trials in adults found no significant increase in serious adverse events with methylphenidate compared to placebo. Similarly, a comprehensive review by Storebø et al. (2018) of 185 randomised trials concluded that whilst methylphenidate may increase the risk of non-serious adverse events, there was no statistically significant increase in serious adverse events compared to placebo or no intervention. However, as most RCTs are of short duration, these findings may not fully capture the risk of psychosis with long-term treatment, a limitation acknowledged in these reviews.

Methylphenidate, a central nervous system (CNS) stimulant, is frequently prescribed for the treatment of ADHD in both children and adults. It is available in immediate-release and extended-release forms, which differ in terms of half-life and duration of action. Although generally considered safe and effective, methylphenidate is associated with common side effects such as insomnia, headaches, loss of appetite and gastrointestinal issues (Storebø et al. 2018). The drug also carries a risk of misuse, particularly amongst adolescents, due to the euphoric effects that can occur at high doses, whether taken orally, intranasally or intravenously (Heal et al. 2013; Shram et al. 2022). A critical distinction that emerges from our review is between the short-term and long-term effects of stimulant medications on psychosis risk. Most randomised controlled trials, including Takahashi et al. (2014), examined relatively short treatment durations (typically 8–12 weeks), which may not capture delayed onset psychotic symptoms or cumulative risk over years of treatment. Conversely, the epidemiological study by Hollis et al. (2019) provided valuable 1-year follow-up data, finding no evidence of increased psychosis risk and observing lower rates of psychotic events in the post-treatment period compared to pre-treatment, though the authors did not conclude that methylphenidate has protective effects. These apparently contradictory findings highlight the possibility of different risk profiles over different time frames: acute adverse events may occur shortly after treatment initiation or dose increases, whilst longer-term dopamine stabilisation might potentially reduce psychosis risk in certain populations. This temporal dimension of risk is particularly important for clinicians to consider, as it suggests the need for closer monitoring during treatment initiation and dose adjustments, whilst possibly offering reassurance about long-term treatment in stable patients. Future research should specifically address this temporal aspect of psychosis risk with stimulant treatment through extended longitudinal studies. Based on available evidence, including literature beyond our primary review, methylphenidate appears to play a significant role in managing ADHD symptoms, with well-established short-term efficacy. Regarding psychotic symptoms, when they do occur in patients taking stimulant medications, case reports and pharmacovigilance data suggest they may often resolve following discontinuation of the stimulant, though the exact proportion requires further investigation through more robust study designs. Current guidelines for ADHD management suggest that cautious reintroduction of stimulants after an episode of stimulant-induced psychosis may be considered in some cases (Cortese et al. 2023), though this approach requires individualised risk assessment.

The comparative study by Moran et al. suggests that methylphenidate may have a lower risk profile than amphetamines regarding psychosis, though the absolute risk of psychotic events appears to be relatively small for both medications (Moran et al. 2019). This finding is consistent with a recent large-scale pharmacoepidemiological study by Solmi et al. (2022) that found amphetamines were associated with a higher risk of psychotic events compared to methylphenidate. Nevertheless, further research is required to explore the outcomes of stimulant re-challenge in individuals with a history of psychotic symptoms. It remains prudent to assess any pre-existing vulnerability to psychosis, as well as the concomitant use of psychostimulant substances before prescribing methylphenidate.

4.1 | Methodological Limitations

We acknowledge several limitations in our methodological approach. Despite the expansion of our search strategy, some relevant studies may have been omitted. In particular, we recognise that the RCT by Takahashi et al. (2014) represents only one of many trials examining stimulant medications, and a more comprehensive approach would involve citing meta-analyses that thoroughly investigate stimulant-related adverse effects, including psychosis. The heterogeneity of included studies limits the possibility of quantitative syntheses. Studies varied considerably in design, population characteristics, outcome measures and

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methodological quality. The lack of high-quality longitudinal studies in some thematic areas restricts our ability to establish causal relationships, particularly regarding long-term effects of stimulant medications. The small number of included studies ($n = 13$) reflects our specific focus on ADHD and first-episode psychosis but may limit the generalisability of our conclusions. Certain conclusions in our original manuscript were based on limited evidence, such as case reports, which we now recognise as a significant limitation. Another important limitation concerns the significant heterogeneity in sample sizes across included studies. Our review encompasses a wide range of sample sizes, from individual case reports to large-scale epidemiological studies involving over 300 000 participants (Moran et al. 2019). Whilst our quality assessment process took sample size into consideration, we acknowledge the inherent challenge in balancing evidence from studies with vastly different statistical power. We attempted to address this by giving greater weight to larger, more robust studies in our synthesis and conclusions, particularly when formulating clinical implications. For instance, the large cohort study by Moran et al. (2019) comparing psychosis risk between methylphenidate and amphetamine users was given more weight in our conclusions regarding pharmacological risk than individual case reports, despite the latter providing valuable insights into clinical presentations and potential mechanisms. We must also acknowledge the potential for publication bias in our review. This is particularly relevant for case reports and small clinical studies, which are more likely to be published when they describe unusual, severe or novel outcomes. This publication bias may lead to an overestimation of adverse events such as psychosis in relation to stimulant treatment. Case reports of psychosis associated with standard doses of stimulants, for example, may be more likely to be published than uneventful clinical courses. Similarly, studies finding no association between ADHD and psychosis risk may be less likely to be published than those identifying significant associations. Whilst we attempted to mitigate this by including larger epidemiological studies and emphasising their findings, the potential for publication bias remains a limitation of the available literature in this field and should be considered when interpreting our conclusions.

5 | Conclusion

This systematic review examined the relationship between ADHD and FEP. Evidence suggests individuals with ADHD, particularly those with neurocognitive deficits or persistent childhood symptoms, may have elevated psychosis risk. Substance use, especially cannabis use, appears to mediate this relationship. Regarding pharmacological treatment, recent epidemiological studies provide a more detailed perspective on methylphenidate safety, with large-scale studies finding no evidence of increased psychosis risk in the immediate period following treatment initiation, though the evidence remains mixed and requires further investigation. When psychotic symptoms occur during stimulant treatment, they often resolve upon discontinuation. Clinicians should consider individual risk profiles, particularly psychosis vulnerability and substance use patterns, when managing ADHD. Further longitudinal research is needed to elucidate these complex relationships.

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The authors have nothing to report.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Quality assessment for randomised controlled trial using Cochrane Risk of Bias Tool 2.0. **Table S2:** Quality assessment for cohort studies using Newcastle–Ottawa Scale. **Table S3:** Quality assessment for case–control studies using Newcastle–Ottawa Scale. **Table S4:** Quality assessment for cross-sectional studies using Newcastle–Ottawa Scale (adapted). **Table S5:** Quality assessment for case series using JBI Critical Appraisal Checklist. **Table S6:** Quality assessment for case reports using JBI Critical Appraisal Checklist.

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